

Both CSVs now train NetMamba+

Packet study · addition to the preserved flow reproduction

1,490,136 rows validated across CICIDS2017 and UNSW CSV exports.

Six native GPU training runs: 1,000 supervised updates each.

A joint model learns from both sources; saved checkpoints support unlabeled inference.

A packet is one message fragment

The paper model originally consumed a sequence from a connection

Original route: five packet byte segments, twenty sizes and twenty intervals.

CSV route: one row with 1,500 byte slots, four metadata values and a label.

Connection identity and reliable sequence order are missing from these exports.

We retain packet inputs; adjacent rows are never invented into flows.

Exactly what goes into the model

Native four-block Mamba encoder, with an explicit packet input contract

1,500 integer bytes → float32 values between -1 and $+1$.

Four bytes per patch → 375 byte tokens + three summary positions.

Size and interval inputs are empty; two prefixes contain no observed flow data.

Native summary pooling → new two-class head → benign / attack logits.

What training actually tests

CIC-only, UNSW-only and joint training; pretrained and scratch for each

Batch 64, seed 0, AdamW; same 1,000-update budget and paired batch streams.

Joint batches contain 32 examples from each dataset.

Pretrained: 50 trunk states transferred, 31 decoder states excluded.

Scratch: fresh weights, matching positional coordinates and starting head.

Validation chooses checkpoints; every selection freezes before test inference.

Measured held-out packet results

Balanced accuracy · each distinct payload has total weight one

Training / initialization	CIC test	UNSW test
cic / pretrained	98.67%	53.73%
cic / scratch	50.00%	50.00%
unsw / pretrained	71.52%	93.22%
unsw / scratch	50.00%	50.00%
joint / pretrained	97.56%	94.29%
joint / scratch	55.57%	50.00%

UNSW-only metadata control: 99.34%. Native NetMamba+ is not universally best here.

Why the data checks matter

Repeated bytes and contradictory labels are retained, not hidden

478,044 distinct payloads across both full exports.

Ten payloads appear in both files; nine have conflicting benign/attack labels.

Identical inputs stay in the same train, validation or test partition globally.

Contradictory groups contribute their observed label proportions to the loss.

CIC test cap: 20,000 groups; UNSW test: 5,930 groups.

Use it and check it

Commands and exact arguments are in `packet-study-setup.md`

Prepare both local CSVs with `packet_data.py`.

Freeze a fresh local protocol, then run `train_packet_model.py`.

Pass an unlabeled payload CSV and packet checkpoint to `predict_packets.py`.

Read `predictions.jsonl` and its fingerprinted completion receipt.

Run `review_packet_study.py` to recompute the published evidence offline.

All 25,930 predicted classes agree; small GPU score differences remain.

What you can defend

Present the measured addition with its limits

Both supplied CSVs genuinely train the native NetMamba+ packet adaptation.

The PDF-guided original flow reproduction remains a separate experiment.

Missing: live capture pipeline, verified flow joins and customer-network validation.

Missing: calibrated packet confidence, NPU export and SmartNIC deployment.

Existing v4 video covers flows and calibration; send this addendum with it.